# Unsaturated hydrocarbons – Reactions of alkenes- Part 2

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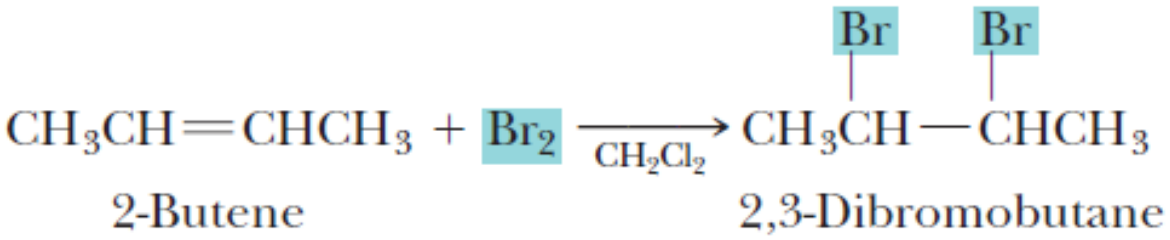
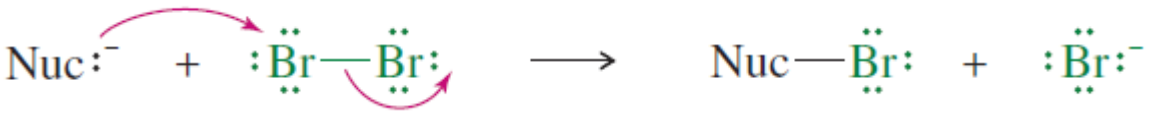
# Halogenation of alkenes

- Chlorine ($Cl_2$) and bromine ($Br_2$) react with alkenes at room temperature by the addition of the halogen atoms to the two carbons of the double bond.

- Two new carbon-halogen bonds are formed.



- A halogen molecule is electrophilic.
- A nucleophile can react with a halogen molecule, displacing a halide ion:

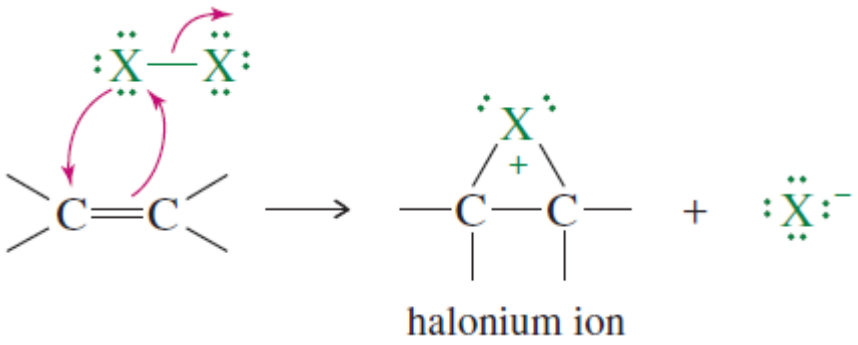



- In this example, the nucleophile attacks the electrophilic nucleus of one bromine atom, and the other bromine atom serves as a leaving group, departing as a bromide ion.

- Similarly, the pi electrons of an alkene can attack the halogen molecule, expelling a halide ion.

Mechanism of alkene halogenation

Step 1:

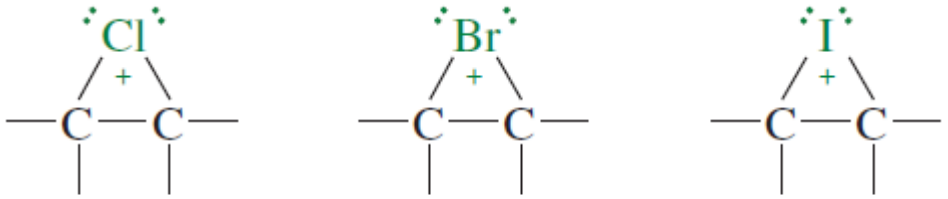



Electrophilic addition of the halogen: the double bond attacks the electrophilic nucleus of one halogen atom forming a new covalent bond with that halogen atom, while the other halogen atom leaves with both electrons from the X—X bond. At the same time, the halogen atom uses its lone pair of electrons to form a bond with the second carbon atom of the double bond.

This results in the formation of a halonium ion.

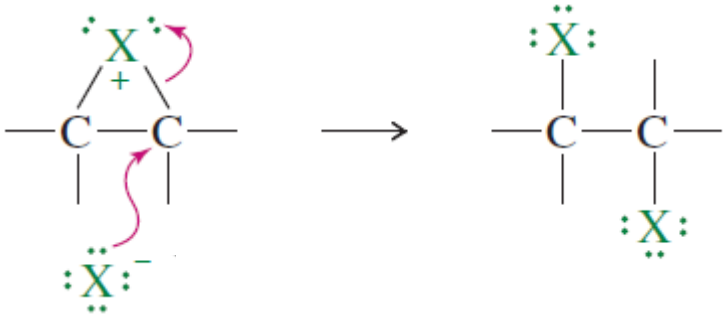
A ***halonium*** ion is a three membered ring with a positive charge on the halogen atom.

Reactions with bromine form bromonium ions, with chlorine form chloronium ions, etc.




chloronium ion bromonium ion iodonium ion

Step 2:

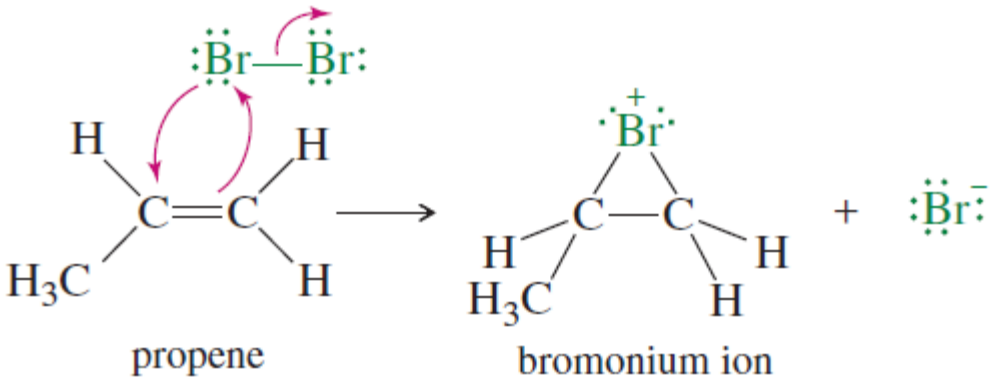



The halonium ion is extremely unstable and highly electrophilic.

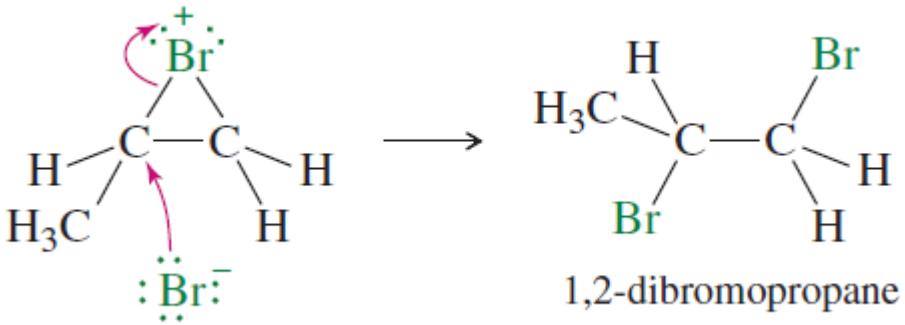
The nucleophilic halide ion, $X^-$, generated in step 1, attacks the electrophilic carbon of the halonium ion, causing the ring to break open to form a stable halogenated product.

- For example, addition of $Br_2$ to propene:

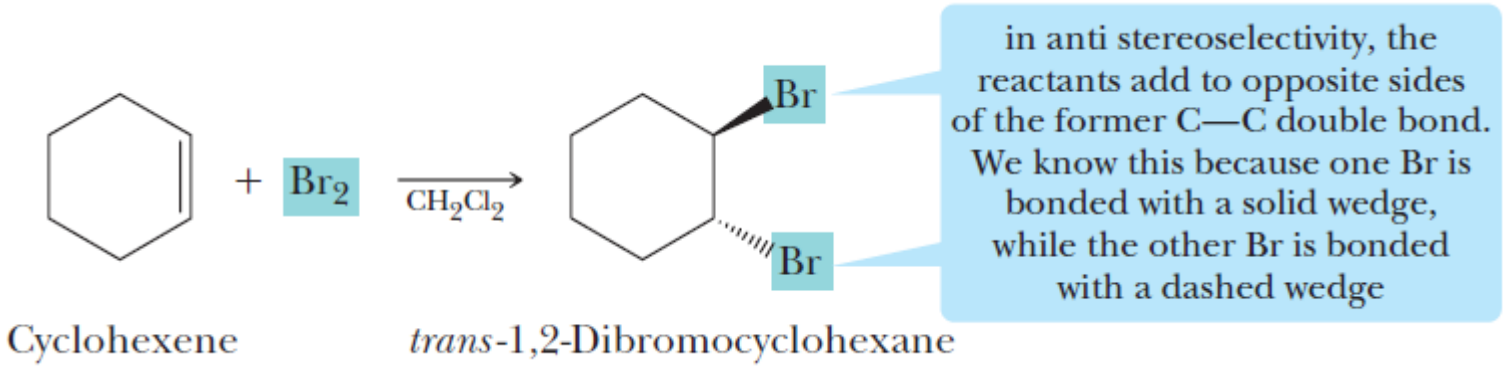
Step 1:




Step 2:

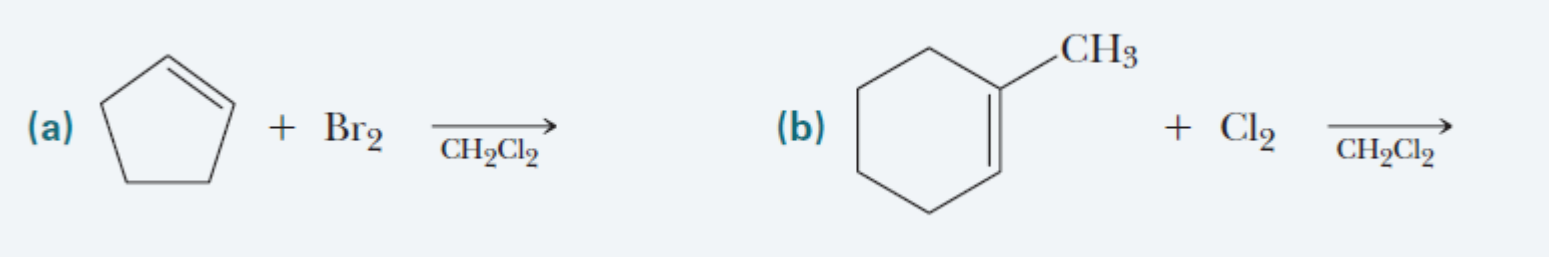


- The addition of halogens across the double bond of cycloalkenes is ***stereoselective***.
- This means one stereoisomer is preferred over the other stereoisomer.
- The addition is ***anti-stereoselective***, or takes place through ***anti-addition***. The halogen atoms add on opposite faces of the double bond, giving rise to the *trans* isomer.



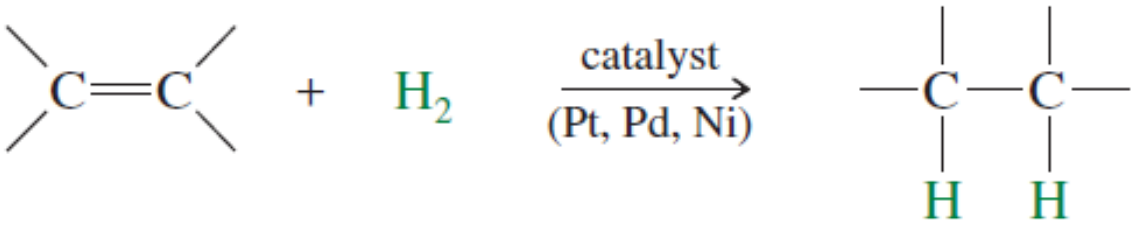
- Practice exercise: Draw the mechanism for the addition of $Br_2$ to 2-butene.

- Complete these reactions, giving the stereochemistry of the products:



# Hydrogenation of Alkenes

- Hydrogen adds across a double bond to give an alkane.
- This is a reduction reaction.
- The process usually requires a catalyst containing Pd, Pt, or Ni.



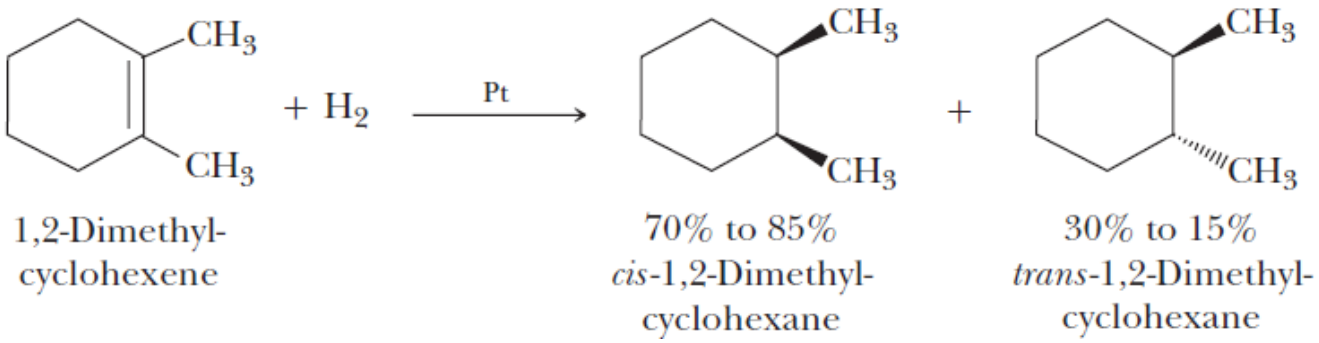

alkene

alkane

*Example*

$$CH_3{-}CH{=}CH{-}CH_3 + H_2 \xrightarrow{Pt} CH_3{-}CH_2{-}CH_2{-}CH_3$$

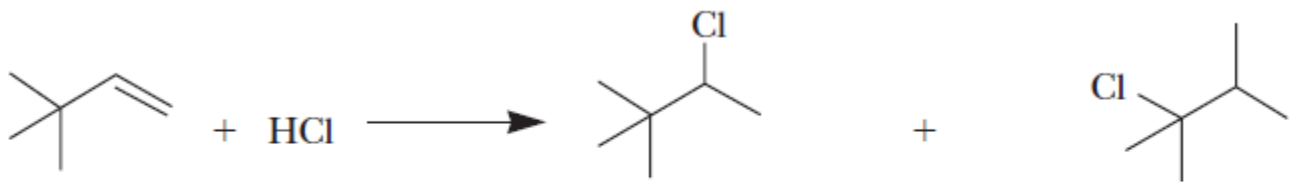
- The reaction is carried out by dissolving the alkene in ethanol or another non-reacting organic solvent, adding the solid catalyst, and exposing the mixture to hydrogen gas at pressures of about 1 – 100 atm.
- Catalytic reduction is ***stereoselective*** (which means the hydrogen adds in a certain manner across a double bond). It takes place through ***syn addition***, where the hydrogens add to the same face across the double bond resulting in a *cis* configuration.



# Carbocation Rearrangements

- In certain cases, migration of a proton or an alkyl group with its bonding pair of electrons, takes place from an adjacent carbon atom to a positively charged carbon atom of a carbocation.
- This is known as a rearrangement.
- This occurs so as to form a more stable carbocation.
- Affects which major product is formed in an addition reaction.

- For example, in the addition of HCl to 3,3-dimethyl-1-butene, only 17% of the expected product 2-chloro-3,3-dimethylbutane is formed.
- The major product instead is 2-chloro-2,3-dimethylbutane.




3,3-Dimethyl-1-butene

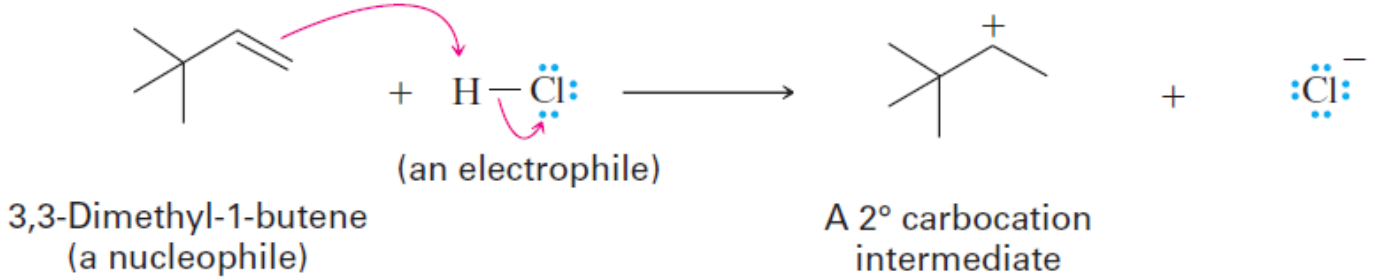
2-Chloro-3,3-dimethylbutane (the expected product 17%)

2-Chloro-2,3-dimethylbutane (the major product 83%)

- Formation of rearranged product can be accounted for by the following mechanism, the key step of which is a type of rearrangement called a **1,2-shift.**

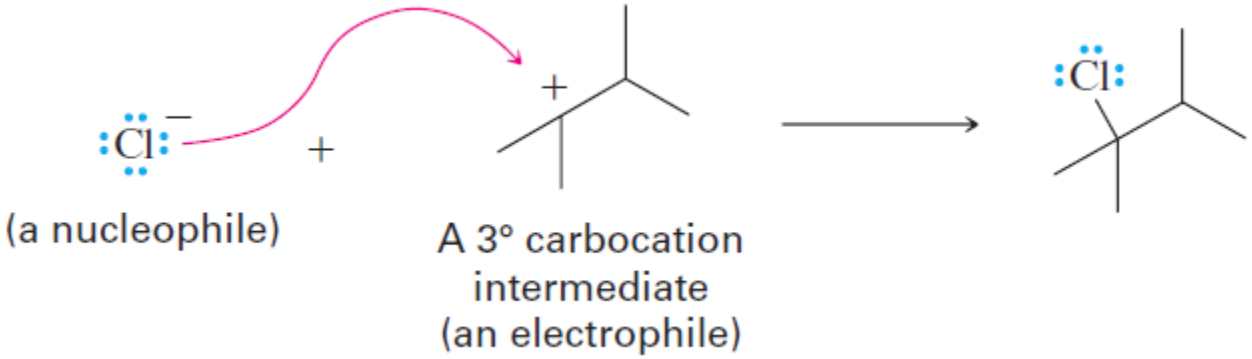
Mechanism of a 1,2-shift:

Step 1: Add a proton: Proton transfer from the HCl (electrophile) to the alkene (nucleophile) gives a 2° carbocation.

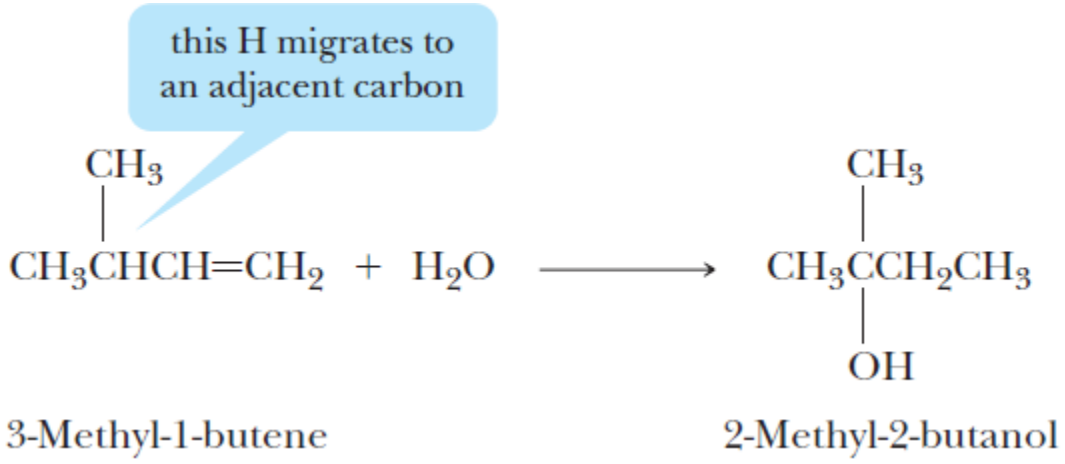



Step 2: Rearrangement of a bond: Migration of a methyl group with its bonding electrons from an adjacent carbon gives a more stable 3° carbocation intermediate. The major movement is that of the bonding electron pair with the methyl group following.

Step 3: Reaction of a nucleophile and electrophile to form a new covalent bond: Reaction of the 3° carbocation intermediate (an electrophile) with chloride ion (a nucleophile) gives the rearranged product.



- The driving force for these rearrangememts are the formation of more stable carbocations.
- The 3° carbocation is more stable than the 2° carbocation.
- Rearrangements also occur in the acid-catalysed hydration of alkenes, especially when a carbocation formed in the first step can rearrange to a more stable carbocation.
- For example, the acid-catalysed hydration of 3-methyl-1-butene gives 2-methyl-2-butanol. In this example, the group that migrates is a hydrogen with its bonding pair of electrons.



- Practice exercise: Propose a mechanism for the acid-catalysed hydration of 3-methyl-1-butene to give 2-methyl-2-butanol.

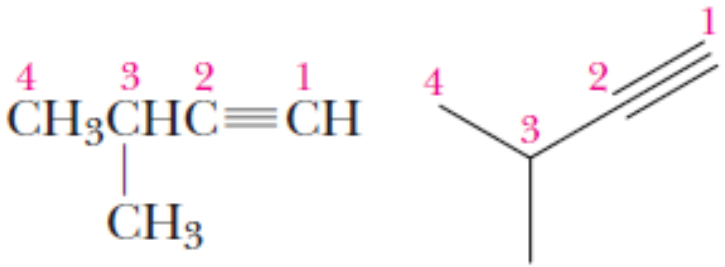
# Alkynes

# Alkynes

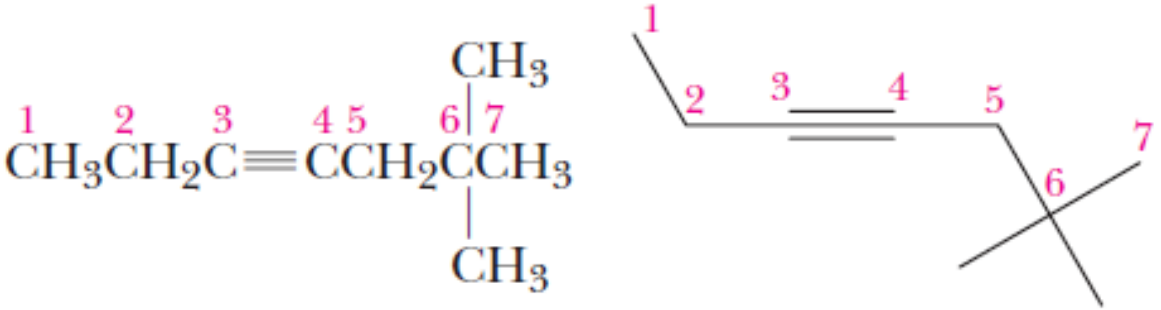
- Alkynes are unsaturated hydrocarbons that have one or more $C \equiv C$ bonds.
- General formula; $C_nH_{2n-2}$
- The simplest alkyne, ethyne, is also known as acetylene.
- Acetylene is commonly used as a fuel in welding torches.
- They are named by identifying the longest continuous chain in the molecule containing a triple bond and adding the ending ‘yne’.
- Same rules as those for naming alkenes.

- The chain is numbered from the end closest to the triple bond, and the position of the triple bond is given by its lower numbered carbon atom.

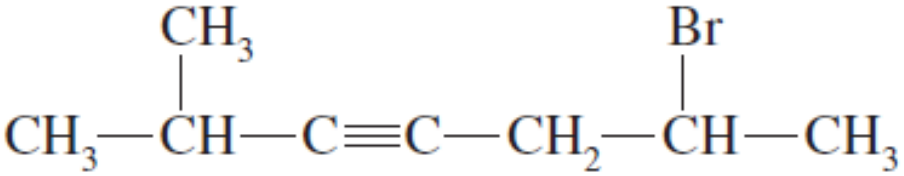
- Substituents are given numbers to indicate their locations.



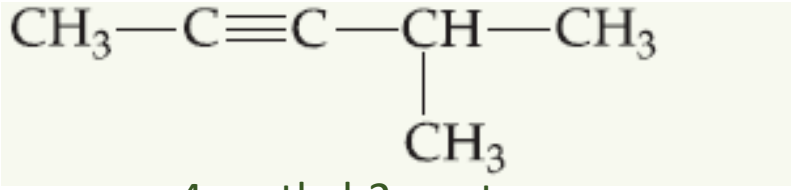
3-Methyl-1-butyne



6,6-Dimethyl-3-heptyne



6-bromo-2-methyl-3-heptyne



4-methyl-2-pentyne

## Physical properties of alkynes:

- They are relatively non-polar, similar to alkanes and alkenes.
- Generally insoluble in water.
- They are quite soluble in organic solvents such as ether, acetone, chloroform, methanol.
- Many alkynes have unpleasant odors.

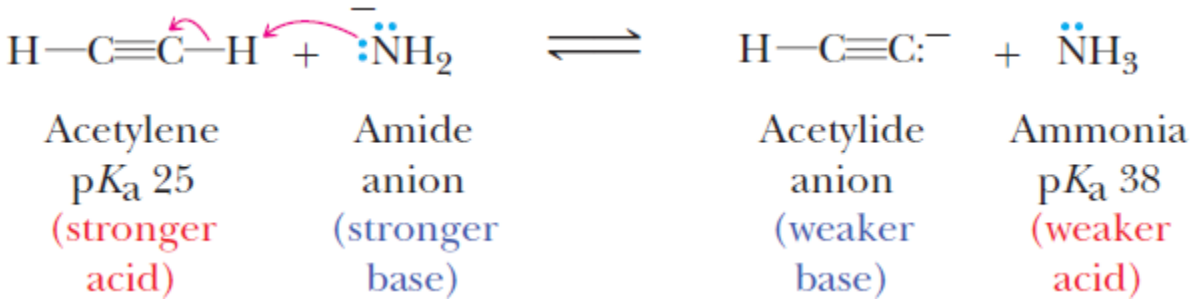
# Reactions of Alkynes

- The hydrogen bonded to a terminal alkyne is relatively acidic enough that it can be deprotonated by a strong enough base such as sodium amide ($NaNH_2$):
- This results in the formation of an acetylide anion.

these hydrogens are much lower in acidity and are not deprotonated by $NaNH_2$

only this hydrogen is acidic enough to be deprotonated by $NaNH_2$

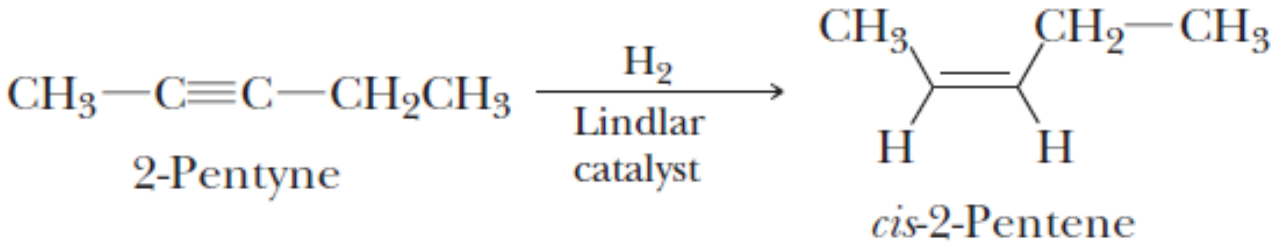
$$CH_3—CH_2—CH_2—C{\equiv}C—H$$



- The acetylide anion is a strong base and a strong nucleophile – it has an unshared pair of electrons that it can donate to an electrophile to form a new carbon—carbon bond.

- Alkynes can be reduced to give alkenes and alkanes.
- Treatment of an alkyne with hydrogen in the presence of a transition metal catalyst, usually Pd, Pt, or Ni, results in the addition of two moles of $H_2$ to the alkyne to form an alkane.

$$\underset{\text{2-Butyne}}{CH_3C{\equiv}CCH_3} + 2H_2 \xrightarrow[\text{3 atm}]{\text{Pd, Pt, or Ni}} \underset{\text{Butane}}{CH_3CH_2CH_2CH_3}$$

- Reaction done at room temperature in the presence of a moderate pressure of $H_2$.

- In the presence of a suitable catalyst, the reaction can be stopped after the addition of one mole of $H_2$, so that the alkyne is converted to an alkene only.
- The catalyst most commonly used for this purpose is **Lindlar's catalyst**.
- **Lindlar's catalyst** consists of finely powdered palladium metal deposited on solid calcium carbonate that has been specially modified with lead salts.
- Reduction of alkynes over Lindlar's catalyst takes place through ***syn addition*** (same face addition) to give a *cis*-alkene.